

Milestone | A/B Testing for H&M

- All of your answers should be typed or pasted into the grey boxes.
- Questions with grey answer boxes expect a specific answer. Once you input the correct answer, the question number box **will change from blue to green**.
- Questions with yellow answer boxes are free-response questions. There will be no change in the question number boxes (in blue), regardless of your answer.
- Hover your cursor over the blue hint box to get a helpful hint!

Part 1: A/B Test Design & Identifying Potential Issues

1	In a sentence or two, state the null hypothesis for H&M's A/B test.
H&M's new subject line does not change the email open rate compared to the original subject line (the open rates are equal). Any observed difference is due to random chance.	
2	In a sentence or two, state the alternative hypothesis for H&M's A/B test.
H&M's new subject line does change the email open rate compared to the original subject line (the open rates are different).	
Use the A/B Test Sample Size Calculator linked below to help you answer the following questions.	
A/B Test Sample Size Calculator	
Historically, promotional emails have had an open rate of 18%. The company wants to see the new subject line increase that to an open rate of 20%.	
3	What is the minimum detectable effect associated with that increase in open rate?
4	What total sample size is required to reliably detect that level of increase?
Before launching any A/B test, it's crucial to identify potential risks or confounding factors that could skew the results. As the analyst, you need to be prepared to be asked "How do we know if this test is valid?"	
5	Use ChatGPT to generate a list of confounding factors by asking "What factors could skew the results of an A/B test like this, and how might they influence our findings or effect the results?" You might also try "What are common mistakes to avoid when designing an A/B test?"
After reviewing ChatGPT's response, identify at least two confounding factors that could impact the experiment. Write a develop a brief strategy (one-to-two sentences) for how you would mitigate these issues if you were a part of designing a similar A/B test.	
Two major confounding factors are uneven audience segmentation and send-time bias. If one group contains more highly engaged customers (or different device users, regions, or prior purchase history), the results may reflect audience differences rather than subject line performance. To mitigate this, we would ensure true random assignment and verify that key audience characteristics are evenly distributed between groups.	

Another confounding factor is timing effects, such as sending one version at a different time of day or during a promotional event. Open rates naturally fluctuate based on timing and external campaigns. To reduce this bias, both

group	subject line	opened
control	20% off + free shipping	0
control	20% off + free shipping	0
control	20% off + free shipping	0
test	Your last chance to score 20% off + free shipping	0
test	Your last chance to score 20% off + free shipping	0
test	Your last chance to score 20% off + free shipping	0
control	20% off + free shipping	0
control	20% off + free shipping	0
test	Your last chance to score 20% off + free shipping	1
test	Your last chance to score 20% off + free shipping	0
test	Your last chance to score 20% off + free shipping	0
test	Your last chance to score 20% off + free shipping	0
control	20% off + free shipping	0
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control	20% off + free shipping	0
test	Your last chance to score 20% off + free shipping	1
test	Your last chance to score 20% off + free shipping	0
test	Your last chance to score 20% off + free shipping	1
control	20% off + free shipping	1
control	20% off + free shipping	0

promotional event. Open rates naturally fluctuate based on timing and external campaigns. To reduce this bias, both versions should be sent simultaneously (or stratified by time zone) and isolated from other concurrent marketing

Part 2: Results Analysis & Interpretation

Data for the A/B Test was collected, and the results can be found in the ABTest data table to the right. In the opened column, 0 indicates that the email was not opened, while 1 indicates the email was opened.

6	In the sheet called PivotTable , create a PivotTable that summarizes the total number of emails sent out to customers in each group, and the number of times each version of the email was opened. It should look like the screenshot!	
7	To check your work, what is the difference in the number of emails <i>sent</i> to each group?	91
8	What is the difference in the number of emails that were <i>opened</i> in each group?	58

Use the A/B Test Significance Calculator below to help you answer the following questions.

[A/B Testing Significance Calculator](#)

9	Which group performed better, the control or the test group? Are your results statistically significant?	
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The test group performed better, with an open rate of approximately 19.13% compared to 17.55% for the control group. The difference is statistically significant at the 95% confidence level ($p \approx 0.047$), so we would reject the null hypothesis and conclude that the new subject line improved email open rates.

10	Since our data is based off of just a sample, it is possible for us to come to an incorrect conclusion. If the truth is in fact different from the conclusion drawn from the data, what type of error would we be making?	
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If the true open rates are actually the same but we concluded that the test subject line performed better, we would be making a Type I error (false positive). This means we incorrectly rejected the null hypothesis when it was actually true.

11	Based on the results of this test, should H&M continue to send promotional emails with the new subject line? Explain why. <i>Remember you can use ChatGPT as your teammate in helping to interpret these results in the context of your hypothesis and the overall goal of increasing email open rate!</i>	
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Yes, H&M should continue using the new subject line because it produced a higher open rate than the control (about 19.13% vs 17.55%), and the difference is statistically significant ($p < 0.05$). This means the lift is unlikely to be due to random chance, so adopting the new subject line supports the goal of increasing email open rates. H&M should still monitor downstream metrics like click-throughs, conversions, and unsubscribes to confirm the higher opens translate into business value.

Part 3: Business Implications

12	If the 'Your last chance to score 20% off + free shipping' subject line increases email open rates but potentially leads to lower engagement or fewer purchases (e.g., customers open the email but don't complete a transaction), how would you balance these factors? What additional data would you need to determine whether the increased open rate translates into meaningful sales?	
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test	Your last chance to score 20% off + free shipping	0
control	20% off + free shipping	0
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control	20% off + free shipping	0
test	Your last chance to score 20% off + free shipping	0
control	20% off + free shipping	0
control	20% off + free shipping	0

would treat purchases/revenue as the primary success metric and use open rate as an early indicator, because higher opens don't matter if they don't lead to transactions. If the new subject line attracts curiosity but lowers intent, I'd balance the tradeoff by optimizing for revenue per recipient (or conversion rate) rather than opens alone, and consider revising the subject line to better match the offer and landing experience. To determine whether the open-rate lift translates into meaningful sales, I'd need data on click-through rate (CTR), conversion rate (purchases ÷ clicks and purchases ÷ recipients), average order value (AOV), revenue per email/recipient, and negative signals like unsubscribe rate, spam complaints, and bounce rate. I'd also compare post-click behavior (cart adds, checkout starts) to see where engagement drops off.

test	Your last chance to score 20% off + free shipping	1
control	20% off + free shipping	0
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